ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Newton's Second Law

Dynamics is force-resultant first, then acceleration.

Newton's Second Law · M1 · 07

eliteigcse.com · WhatsApp +20 112 000 9622

Dr Eslam Ahmed · Cairo University Faculty of Engineering

The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 NET: Net Force
TRIGGER *Newton's second law*

02 TENS: Connected Tension
TRIGGER *connected particles, string*

03 SYS: Whole System
TRIGGER *find acceleration of system*

04 COMBO: Motion Plus Dynamics
TRIGGER *force then distance/time*

BANK EVIDENCE Local pages: 49 problem, 78 answer, 25 notes. Website primary entries: 12. Website linked entries: 55.

01 NET: Net Force

TRIGGER *Newton's second law*

BECOMES Resultant force in a direction equals mass times acceleration.

FIRST LINE TO WRITE

$$\sum F = ma$$

SIMPLEST STRATEGY

1. **N**ame the direction.
2. **E**nter signed forces.
3. **T**ie to ma.

WORKED MODEL

If right is positive, $P - F = ma$.

HARD VARIANTS

1. Use weight mg, not mass, in vertical force sums.
2. Acceleration sign must match the chosen positive direction.
3. If acceleration is zero, this becomes equilibrium.

BOTTOM LINE

Only the resultant force equals ma.

Practice Page: NET

QUESTION TYPE Newton's law

Find acceleration when a 6 kg particle is pulled by 30 N against 6 N resistance.

COMMON TRAP Writing one force equal to ma.

02 TENS: Connected Tension

TRIGGER *connected particles, string*

BECOMES Use common acceleration and separate body equations.

FIRST LINE TO WRITE

$$a_A = a_B, \quad \text{same string}$$

SIMPLEST STRATEGY

1. **T**race the string.
2. **E**stablish common acceleration.
3. **N**ame tension.
4. **S**et one equation per body.

WORKED MODEL

For two particles: $P - T = m_1 a$, $T - R = m_2 a$.

HARD VARIANTS

1. Light string gives the same tension throughout that string.
2. Use whole-system equation to find acceleration quickly, then individual equation for tension.
3. Pulley or slope geometry may reverse signs.

BOTTOM LINE

Tension is internal to the system but external to each particle.

Practice Page: TENS

QUESTION TYPE Connected particles

Two connected particles move on a smooth table; find T and a.

COMMON TRAP Using different accelerations for an inextensible string.

03 SYS: Whole System

TRIGGER *find acceleration of system*

BECOMES Add bodies together to cancel internal tensions.

FIRST LINE TO WRITE

$$\text{external resultant} = (m_1 + m_2)a$$

SIMPLEST STRATEGY

1. **S**elect the whole moving system.
2. **Y**ield only external forces.
3. **S**olve for common acceleration.

WORKED MODEL

> If two masses move together, $P - R = (m_1 + m_2)a$.

HARD VARIANTS

1. Only include bodies that share the same acceleration.
2. External resistance and weight components remain.
3. Return to a single-body equation for tension.

BOTTOM LINE

Internal tensions cancel in the whole-system equation.

Practice Page: SYS

QUESTION TYPE System dynamics

Find the acceleration of two connected particles using the whole system.

COMMON TRAP Including tension when treating the whole system.

04 COMBO: Motion Plus Dynamics

TRIGGER *force then distance/time*

BECOMES Find acceleration from forces, then use kinematics.

FIRST LINE TO WRITE

$$\sum F = ma \quad \Rightarrow \quad \text{SUVAT}$$

SIMPLEST STRATEGY

1. **C**alculate acceleration.
2. **O**rganise initial motion data.
3. **M**atch SUVAT equation.
4. **B**ring units together.
5. **O**utput requested time/speed/distance.

WORKED MODEL

$a = 2$, then $v^2 = u^2 + 2as$.

HARD VARIANTS

1. If a force stops acting, split the motion into stages.
2. Friction may change direction after motion reverses.
3. Use the acceleration from the correct stage only.

BOTTOM LINE

Dynamics often supplies the missing SUVAT acceleration.

Practice Page: COMBO

QUESTION TYPE Dynamics plus kinematics

A pulled particle starts from rest; find its speed after moving 10 m.

COMMON TRAP Using force equations after acceleration changes.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
Newton's second law	$\sum F = ma$
connected particles, string	$a_A = a_B$, same string
find acceleration of system	external resultant $= (m_1 + m_2)a$
force then distance/time	$\sum F = ma \quad \Rightarrow \quad$ SUVAT

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
NET: Net Force	Use weight mg, not mass, in vertical force sums. Acceleration sign must match the chosen positive direction. If acceleration is zero, this becomes equilibrium.
TENS: Connected Tension	Light string gives the same tension throughout that string. Use whole-system equation to find acceleration quickly, then individual equation for tension. Pulley or slope geometry may reverse signs.
SYS: Whole System	Only include bodies that share the same acceleration. External resistance and weight components remain. Return to a single-body equation for tension.
COMBO: Motion Plus Dynamics	If a force stops acting, split the motion into stages. Friction may change direction after motion reverses. Use the acceleration from the correct stage only.